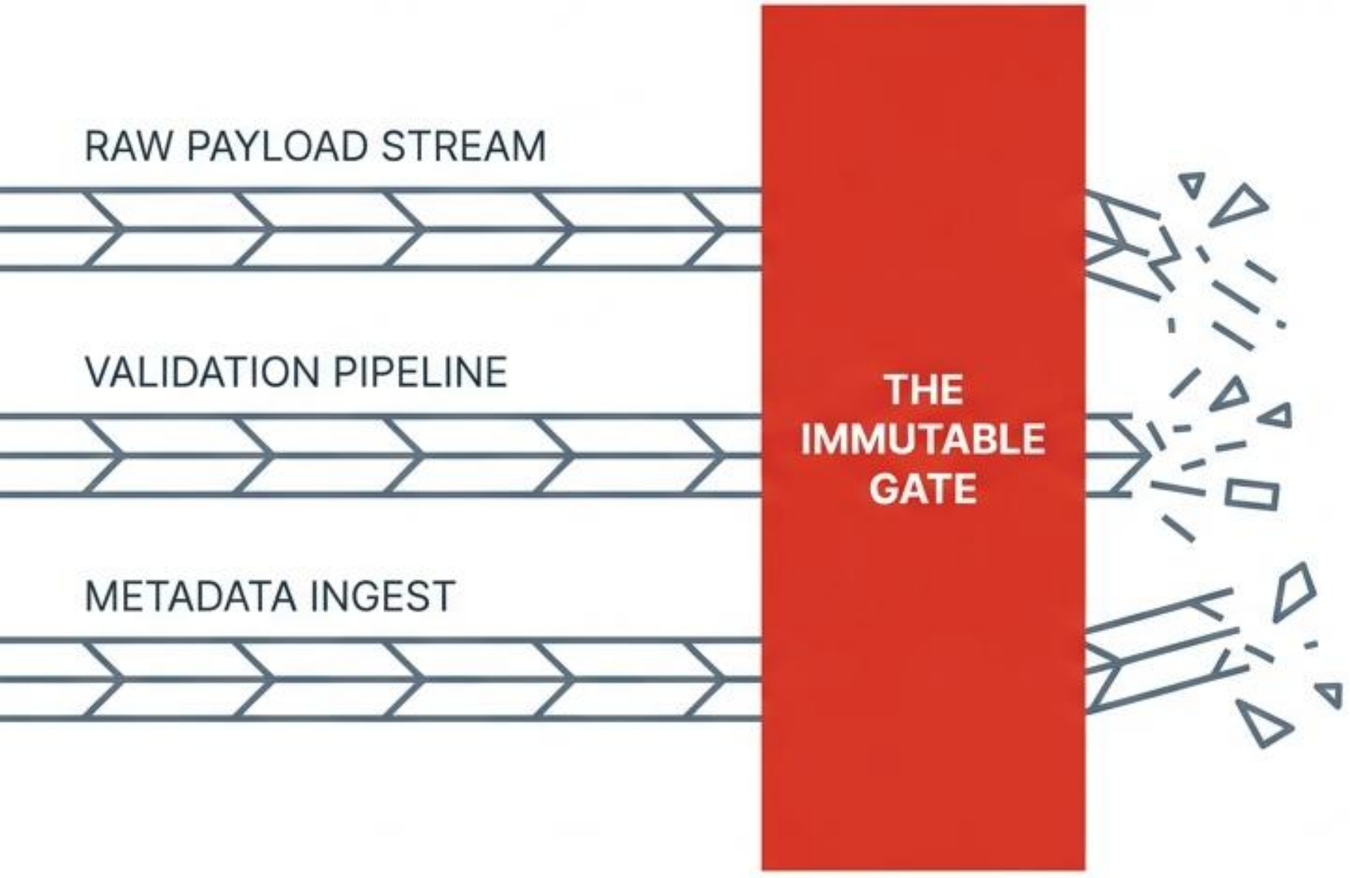
The Webhook Interceptor — stopping bad data at the door.

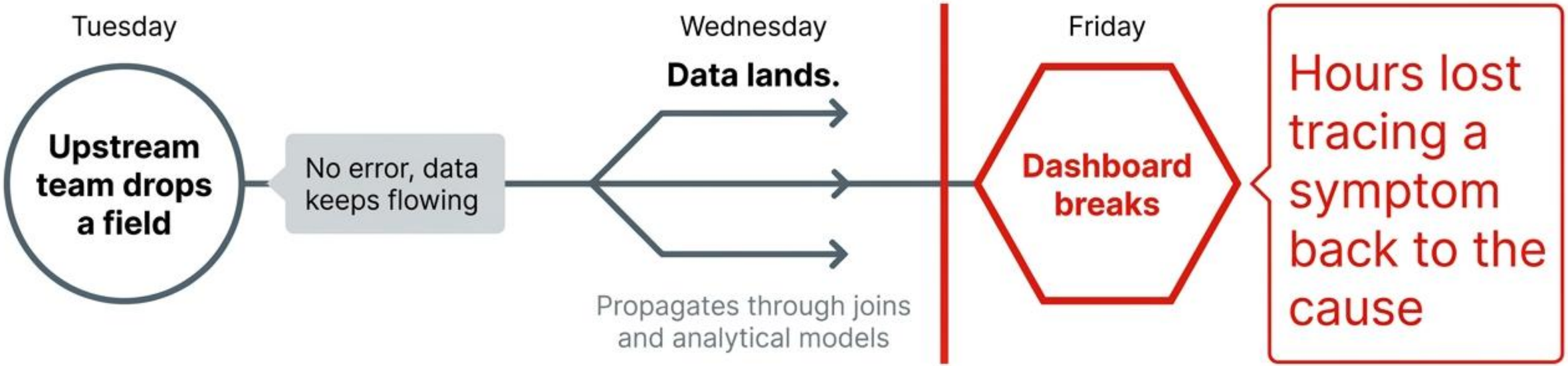
Project 1 of 3 · a shift-left data-engineering portfolio



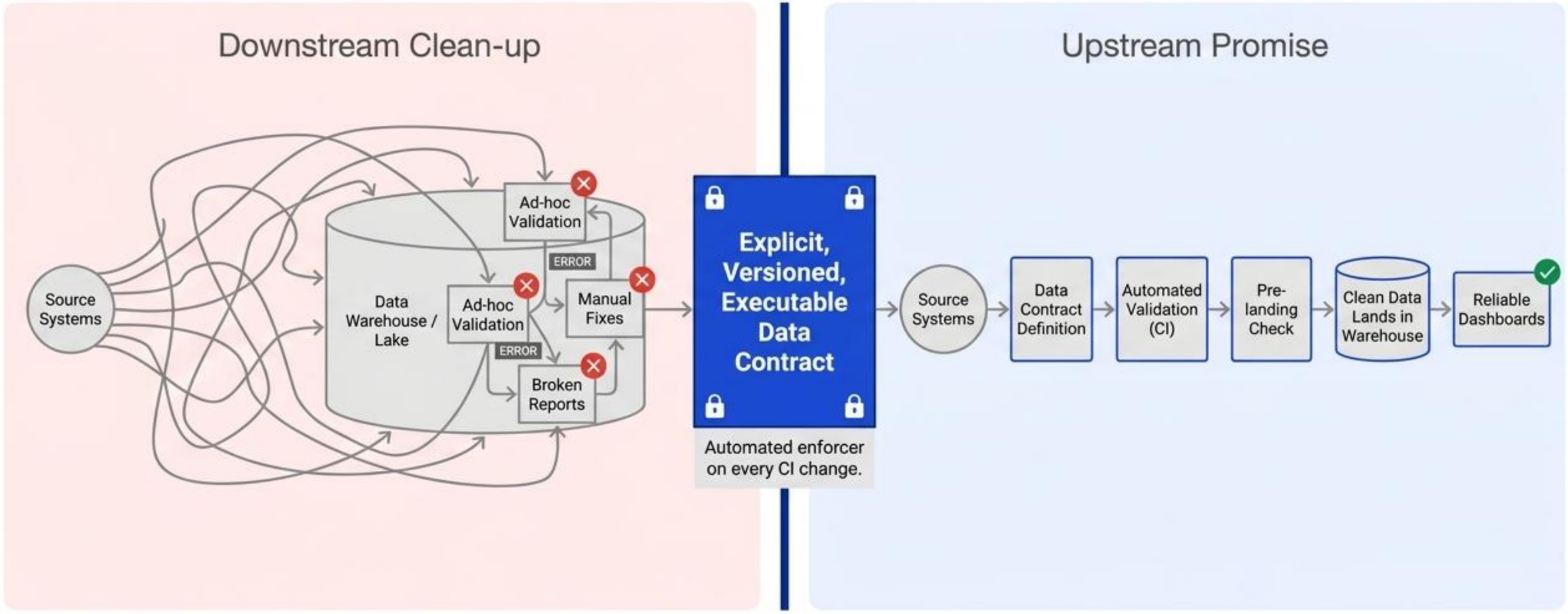
Runnable in ~2 minutes · synthetic data · free/local tooling

Silent breaking changes are found downstream, days late.

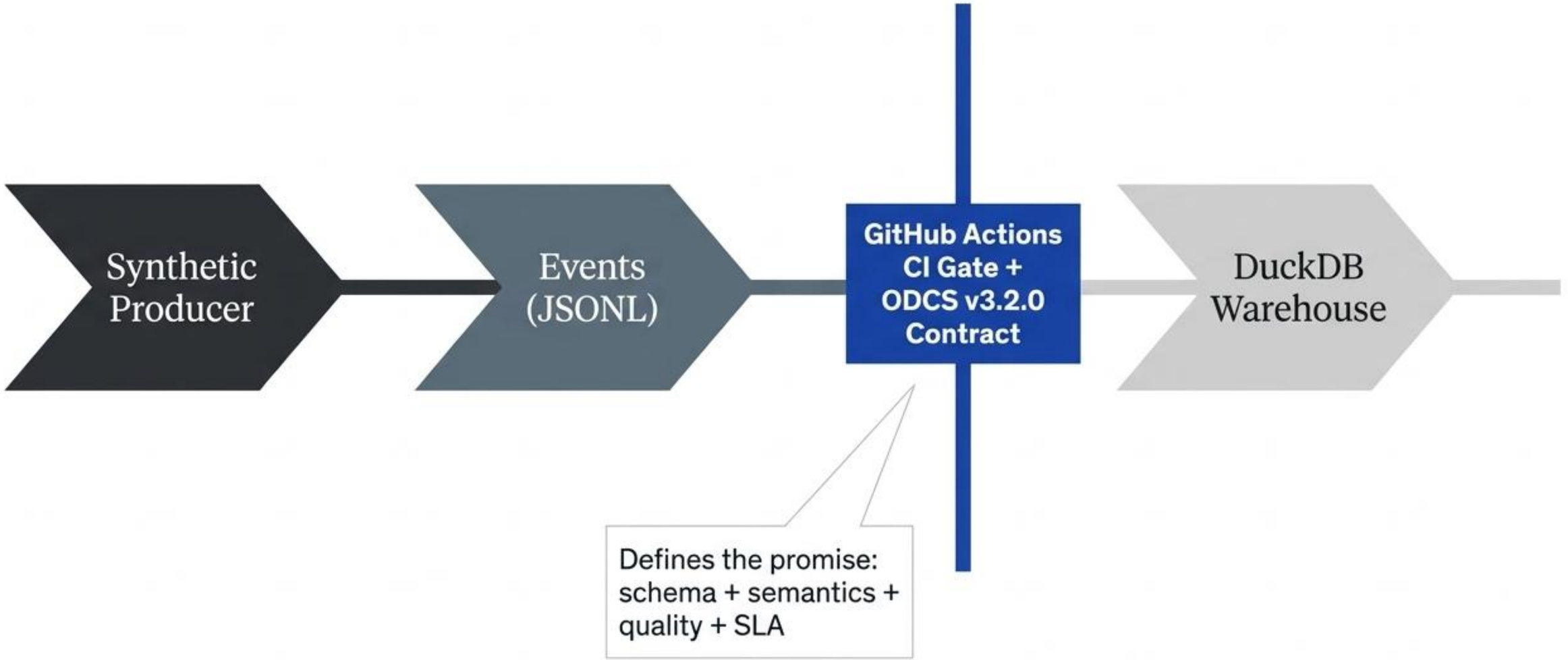
The Tuesday-to-Friday Lag | The Tuesday-to-Friday Lag



Data quality is an upstream promise, not a downstream clean-up.



A deliberately small pipeline, so the mechanism is unmistakable.



An ODCS contract is more than a schema.

Semantics

Business Context — Captures business names and descriptions for every field, ensuring “amount” and “currency” hold business meaning, not just data types.

Quality Rules

Data Integrity — Enforces transaction_id as unique/non-null, amount within range, and currency restricted to an ISO-4217 enum.

Classification

Governance — Tags account_id as restricted (creating the hook for downstream privacy mapping).

SLA

Expectations — Defines freshness and completeness guarantees.

Break it on purpose → CI goes red → the merge is blocked.

```
make demo
```



Valid data flows through.
Contract test passes.

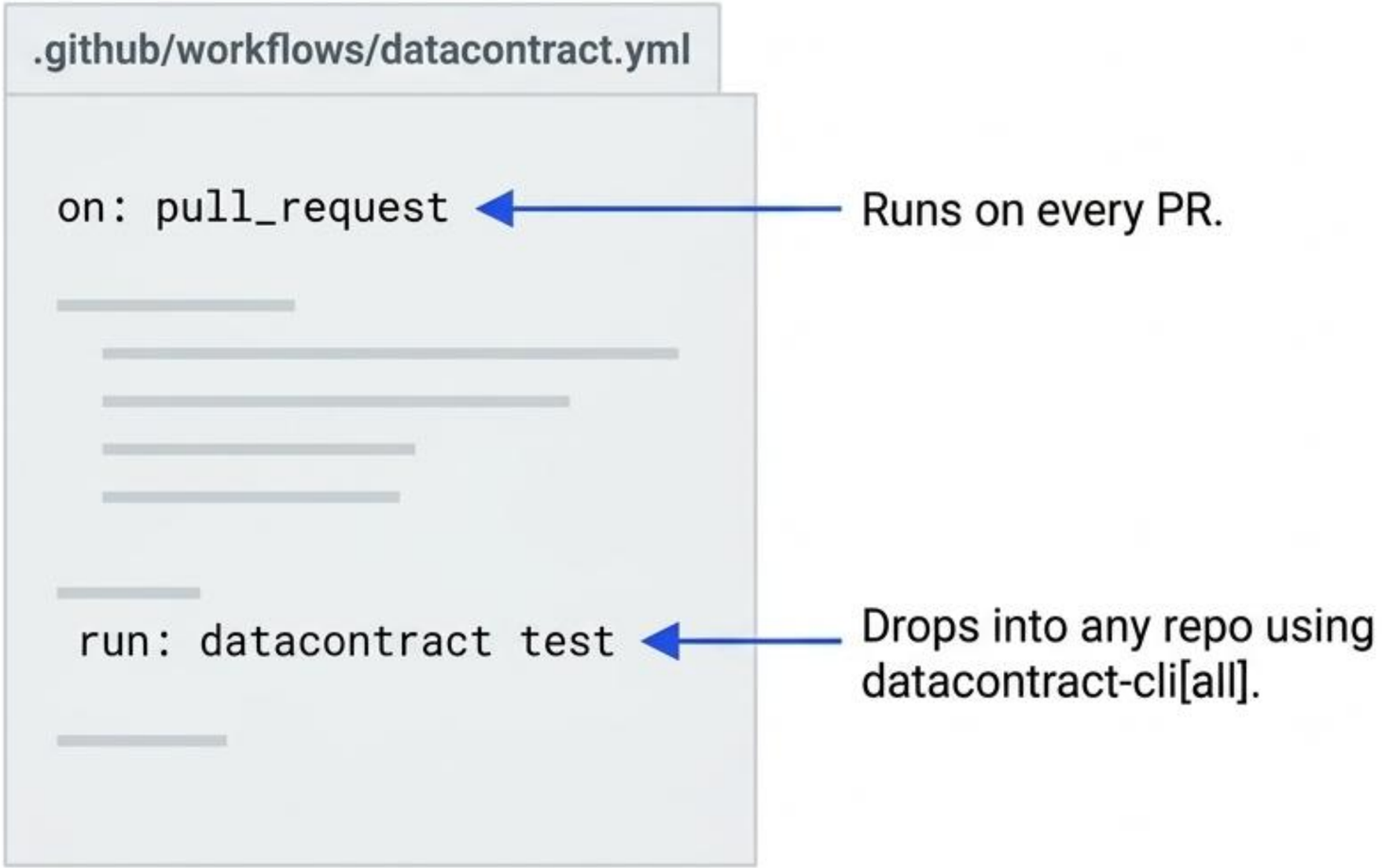
Producer drops the required currency field.

```
make demo-break
```



datacontract test fails; pull request blocked before data reaches production.

Contract enforcement is a few lines of GitHub Actions.



The architecture of the gate: The contract does the thinking. CI runs it where it can block a merge.

Choices that make it credible and reproducible.

Engineering Decision	Strategic Rationale
DuckDB (local)	Zero-config; anyone can run it instantly without a cloud account.
Synthetic data generator	Fully reproducible; guarantees no real or PII data is used.
Pinned tooling	Precision execution; locked to **datacontract-cli[all] 1.2.7** ensuring ODCS **v3.2.0** compatibility.
First-class enum on currency	Future-proof; aligned tightly with current ISO-4217 standards.

The same control is operational-resilience evidence (CPS 230).

Technical Execution

Technical SLA isolating critical operations from upstream disruption. **(Runs in CI on every change).**

Translates to

CPS 230 Evidence

A continuous, auditable integrity trail showing implementation of controls **(not just point-in-time attestations).**

General information only, not compliance advice. CPS 230 is fully in force; supervision focuses on evidence of effective implementation.

What’s simplified, and why.

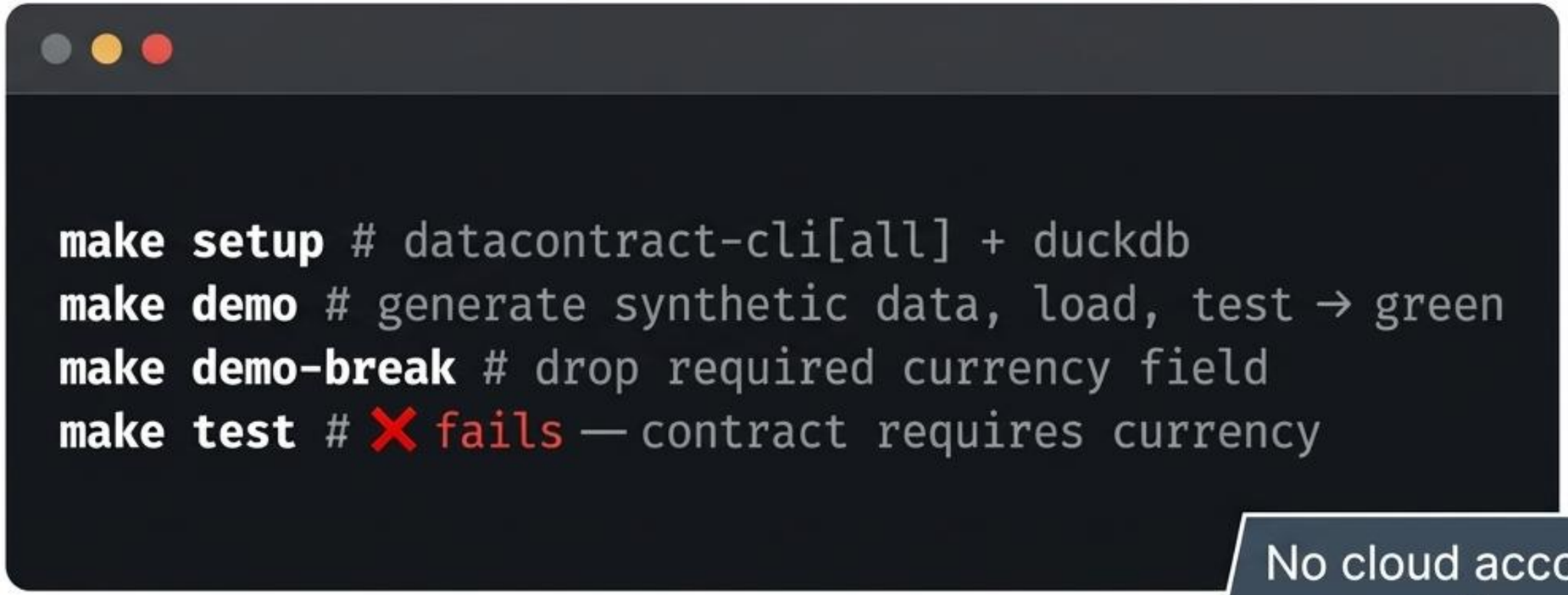
Demo Simplifications

The webhook is a local JSONL file (not a live Kafka stream). DuckDB stands in for Snowflake/BigQuery. Small data volume.

Production-Grade Core

The exact mechanism — contract-enforced validation blocking breaking changes in CI — is exactly what you would run in enterprise production.

Clone, run, break it — in about two minutes.



No cloud account,
no credit card.

What this project proves I can do.

Data Governance

Author real ODCS v3.2.0 data contracts mapping semantics, quality, SLA, and classification.

Platform Engineering

Build CI/CD data-quality gates utilizing GitHub Actions and datacontract-cli.

Data Engineering

End-to-end Python ingestion and local DuckDB warehousing.

Risk & Strategy

Map concrete engineering controls to prudential regulations (APRA CPS 230).

See the code · connect

Repo: github.com/initiative1972/shift-left-1-webhook-interceptor

Author: Henry Yan · AI Engineer & Data Architect · Melbourne

Scan to view repository

Portfolio Sequence Tracker

Project 1 (prevent) — Project 2 (enforce) — Project 3 (prove)

Personal project · synthetic data only · general information, not advice · views my own.